

**VIT**

Vellore Institute of Technology

Final Assessment Test – April 2025

Course: BCSE304L - Theory of Computation

Class NBR(s): 1613/1615/1617/1619/1621/1623/1625/

1627/1629/1631/1633/1635/1637/1639/1641/1643/

1645/1647/1649/1651/1653/1655

Slot: F1+TF1

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1.
 - a) Let $L_1 = \{ n^2 \% 3 = 0 \text{ and } n \in \mathbb{Z} \}$, $L_2 = \{ n \% 3 = 0 \text{ and } n \in \mathbb{Z} \}$. (Note: % is modulo operator). Prove by contrapositive, "If L_1 then L_2 ". [6]
 - b) Let $\Sigma = \{ \#, \$, \& \}$ be an alphabet. How many distinct strings of length 5 can be formed over Σ . [2]
 - c) Let $\Sigma = \{a, b\}$ and consider the language [2]

$$L_1 = \{ w \in \Sigma^* \mid w \text{ begins with } a \text{ and } |w| \geq 3 \}$$
 and

$$L_1 - L_2 = \{aba, abb\}$$
 Determine one possible definition for the language L_2 (using set builder notation or formal description) such that $L_1 - L_2$ is satisfied.
2.
 - a) Write a Regular Expression for a language that cannot have two consecutive a's together over $\Sigma = \{a, b\}$ [3]
 - b) Based on the above regular expression which you obtained in the Question No. 2. (a), convert it into Non-Deterministic Finite Automata. [7]
3. Convert the following Context Free Grammar (CFG) into an equivalent Greibach Normal Form (GNF). Also prove the string "abab" is derivable in 4 steps of Right Most Derivation via the resultant GNF.

$$S \rightarrow SS \mid aSb \mid ab$$

S is the start symbol of the CFG.

4. Design a Push Down Automata (PDA) for
 $L = \{ a^i b^j c^k d^l \mid i=j \text{ or } j=l, i,j,k,l > 0 \}$ and justify whether your obtained PDA is deterministic (DPDA) or non-deterministic (NPDA)?

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5. Design a Turing Machine (TM) that performs the following operation over $\Sigma = \{0,1\}$. Also briefly discuss about your idea for constructing transition rules and explain how you process example 3 or example 4.

Copy Paste operation

For Example 1 – Input : 0011

Output : 00110011

Example 2 – Input: 010010

Output: 010010010010

Example 3 – Input: 00

Output: 0000

Example 4 – Input: 111

Output : 111111

6. Either Prove or Disprove the following statements: [3+4
+3]
- i. If L and $\bar{L}$ are Recursively Enumerable, then both languages are recursive
 - ii. If L is recursive, then $\bar{L}$ is also recursive.
 - iii. If L and $\bar{L}$ are Recursive, then both languages are recursively Enumerable.
7. a) A student X derives a string (w) of length K from Chomsky Normal Form (CNF). [3]
 An another student Y derives the same string (w) of length K via Greibach Normal Form (GNF). Who would derive the string w quickly? X or Y? Justify.
- b) Let $L = \{w.w^R \mid w \in \{a,b\}^+\}$ is a context Free Language (CFL). (Note: w^R means w reversed). [7]
 Answer the following with proper justification

OR

9.b) Design a Turing Machine for unary to ternary (Base 3) convertor.

For Example 1 - Input: 00000

Output : 12

Example 2 - Input : 000

Output : 10

Example 3 - Input : 000000000000000

Output: 120

10.a) Let $L1 = \{ w \mid \text{length of } w \text{ is not even} \}$ and $L2 = \{ w \mid \text{length of } w \text{ is divisible by 3} \}$ over $\Sigma = \{0,1\}$. Let $M1$ and $M2$ are the corresponding Deterministic Finite Automata (DFA) for $L1$ and $L2$.

Find

- $M1 \times M2$ (means cross product of $M1$ and $M2$. That means DFA accepting $L1 \cap L2$).
- Complementation of $M1 \times M2$.
- Reversal of $M1 \times M2$.

OR

10.b) i) Let $M = (Q, \Sigma, \delta, q_0, F)$ be a Deterministic Finite Automata (DFA) [5]
Statement 1 : If you reverse M , the result automata (M^R) is either DFA or Non-Deterministic Finite Automata (NFA)

Statement 2: If you complement a DFA, the resultant language is also complemented.

Statement 3: If you complement a NFA, the resultant language need to be complemented always.

Analyse the truthness of Statement 1, 2 and 3 and justify your answer.

ii) Let us assume the length of a string is n . If so, how many states are required [5]
for minimized DFA that accepts the following:

- Exactly n length strings
- Atleast n length strings
- Atmost n length strings

⇔⇔⇔ F/E/TY ⇔⇔⇔

Reg. No: 22BC10018



Final Assessment Test – May 2024

Course: BCSE304L - Theory of Computation

Class NBR(s): 0760/0763/0766/0768/0771/0774/

0784/0789/0797/0845/0861/1014/1015/1026/

1029/1036/1037/1042/1043/1045/1047/1049/

1051

Slot: B2+TB2

Time: Three Hours

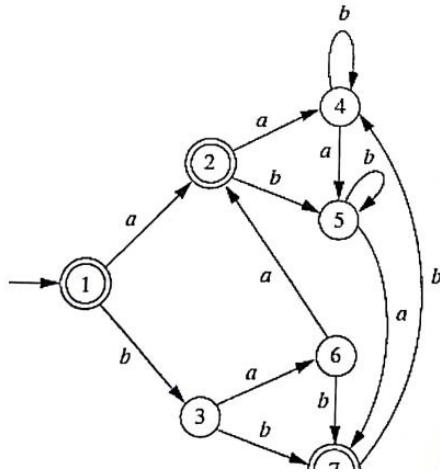
Max. Marks: 100

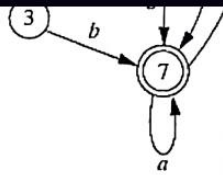
- KEEPING MOBILE PHONE/ELECTRONIC DEVICES EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
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Answer any TEN Questions

(10 X 10 = 100 Marks)

1. a) Construct a Context Free Grammar for the following languages: [6]
 - i) $L = \{ a^m b^n \mid n = 2m, m, n \geq 1 \}$
 - ii) $L = \{ a^n b^m c^k \mid k = |n-m|, n, m \geq 1 \}$
- b) Use induction on n to show that $|u^n| = n |u|$ for all strings u and all n . [4]
2. a) Construct NFA over the alphabet $\Sigma = \{a, b\}$ which accepts all strings whose length is divisible by 5. [5]
- b) For the constructed NFA in Question 2a, construct an equivalent DFA. [5]
3. Construct a Minimized DFA using Table Filling Method





4. Consider the regular expression
 $R = (aa)^* + ba^* + (ab)^*$
 - a) Construct an NFA for the above regular expression. [5]
 - b) Construct a regular grammar G, for the language generated by the regular expression $aa^* (ab/a)^*$. [5]
5. Construct a DFA for the following language over $\{a,b\}$.
 $L = \{w \mid w \text{ does not contain substring } aa \text{ or } bb\}$.
 From the obtained DFA, use any appropriate method and derive the corresponding regular expression.
6. Determine whether the string $w = \text{He eats a fish with a fork}$ is in the language generated by the given grammar G, using the CYK algorithm.

$G = \{$
 $S \rightarrow NP VP$
 $VP \rightarrow VP PP \mid V NP \mid \text{eats}$
 $PP \rightarrow P NP$
 $NP \rightarrow \text{Det } N \mid \text{He}$
 $V \rightarrow \text{eats}$
 $P \rightarrow \text{with}$
 $N \rightarrow \text{fish} \mid \text{fork}$

with

$N \rightarrow fish \mid fork$

$Det \rightarrow a$

}

7. a) Find an equivalent grammar in CNF for the given grammar. [6]

$G = (\{S, A, B\}, \{a, b\}, \{S \rightarrow ASB \mid \epsilon, A \rightarrow aAS \mid a, B \rightarrow SbS \mid A \mid bb\}, S)$

- b) Check whether the following grammar is ambiguous or not. Show the derivation steps. K is the start symbol. [4]

$K \rightarrow QK \mid \epsilon, Q \rightarrow Qa \mid aQb \mid ab$

8. Construct a Pushdown Automata (PDA) that accepts the language $L = \{a^n b^{m+n} c^m : n \geq 0, m \geq 1\}$ by empty stack method. Give the Instantaneous description for the string $aaabbbbcc$.

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9. Design a Turing Machine transducer that takes the input $w \in \{a, b\}^*$ and produces a copy of it as an output. For example, if $w = aba$, then the output will be $abaaba$.

10. a) For an Universal Turing machine, obtain the code for [6]

$\langle M, 1010 \rangle$ where

$M = (\{q_1, q_2, q_3\}, \{0, 1\}, \{0, 1, B\}, \delta, q, B, \{q_2\})$ have moves

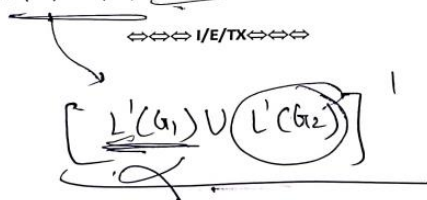
$\delta(q_1, 1) = (q_3, 0, R)$

$\delta(q_3, 0) = (q_1, 1, R)$

$\delta(q_3, 1) = (q_2, 0, R)$

$\delta(q_3, B) = (q_3, 1, L)$

9. Design a Turing Machine transducer that takes the input $w \in \{a,b\}^*$ and produces a copy of it as an output. For example, if $w=aba$, then the output will be $abaaba$.
10. a) For an Universal Turing machine, obtain the code for $\langle M, 1010 \rangle$ where $M = (\{q_1, q_2, q_3\}, \{0,1\}, \{0, 1, B\}, \delta, q, B, \{q_2\})$ have moves
- $\delta(q_1, 1) = (q_3, 0, R)$
 $\delta(q_3, 0) = (q_1, 1, R)$
 $\delta(q_3, 1) = (q_2, 0, R)$
 $\delta(q_3, B) = (q_3, 1, L)$
- b) Does PCP with two lists $x=\{b, bab3, ba\}$ and $y=\{b3, ba, a\}$ have a solution? Justify. [4]
11. Construct a Turing machine that will accept the language $L = \{a^n b^m a^{n+m} : n \geq 0, m \geq 1\}$ over the alphabet $\Sigma = \{a, b\}$.
12. a) Discuss recursive languages and recursively enumerable languages. [4]
 b) Let G_1 be any unrestricted grammar, and G_2 be any regular grammar. Show that the problem $L(G_1) \cap L(G_2) = \emptyset$ is undecidable. [6]





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Final Assessment Test – May 2024

Course: BCSE304L - Theory of Computation

Class NBR(s): 0758/ 0762/0764/0767/0769/0770/

0773/0783/0788/0794/0842/0859/1011/1013/1024/ Slot: B1+TB1

1027/1028/1031/1034/1038/1040

Time: Three Hours

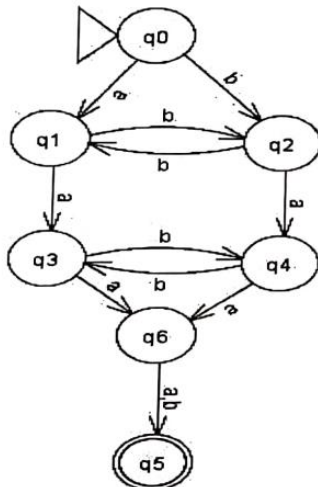
Max. Marks: 100

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Answer any **TEN** Questions

(10 X 10 = 100 Marks)

1. a. If $x+y$ is an even number, then prove that x and y are of same parity [4]
(even-even or odd-odd) using contra-positive.
- b. $L_1 = a^n b^m$ and $L_2 = b^m$, where $n, m = 2$ find $(L_1 \cap L_2)^2$ and $(L_1 \cup L_2)^2$ [6]
2. Minimize the following DFA using table filling method



3. Convert the following NFA with epsilon moves to an equivalent NFA without epsilon moves where the starting state is q_0 and final state is q_2 .

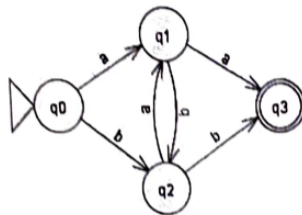
States/Input	ϵ	a	b	c
q_0	q_1			

3. Convert the following NFA with epsilon moves to an equivalent NFA without epsilon moves where the starting state is q_0 and final state is q_2 .

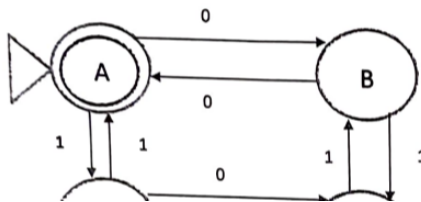
States/input	ϵ	a	b	c
q_0	Φ	$\{q_0\}$	$\{q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$	$\{q_2\}$	Φ
q_2	$\{q_1\}$	$\{q_2\}$	Φ	$\{q_0\}$

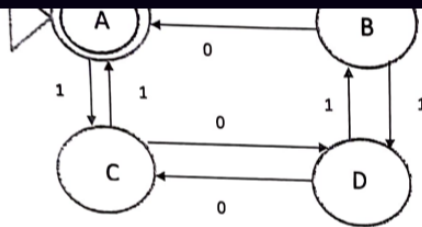
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4. Derive the regular expression from the below DFA using Ardens and State elimination method.



5. a) Convert the given finite state automata accepting the language [5]
 $L = \{w \mid n_0(w) \bmod 2 = 0 \text{ and } n_1(w) \bmod 2 = 0\}$ where $n_0(w)$ denotes the number of 0's in w and $n_1(w)$ denotes the number 1's in w to an equivalent regular grammar. Where the start state is A.





b) By using pumping lemma, prove that the following language is not regular [5]
 $L = \{a^n b^l a^k \mid n=l \text{ or } l \neq k\}$

6. a) Prove the context free Grammar, G is having ambiguity for the string [5]
 $w = aaaabbbb$, (S is the start symbol).
 $S \rightarrow ASB \mid BSA \mid AB \mid BA$
 $A \rightarrow Aa \mid a$
 $B \rightarrow Bb \mid b$

b) Convert the given context free grammar G, into an equivalent context free [5]
 grammar G' in Chomsky Normal Form (CNF), where the start symbol is A.
 $A \rightarrow BAZ$
 $B \rightarrow aBA \mid a \mid \epsilon$
 $Z \rightarrow AbA \mid B \mid bb$

7. Check the membership for the string $w = babab$ by using CYK algorithm for the given grammar, where S is the start symbol.

$S \rightarrow AC \mid CB$

$A \rightarrow CA \mid a$

$B \rightarrow AB \mid b$

$C \rightarrow CB \mid b$

7. Check the membership for the string $w = babab$ by using CYK algorithm for the given grammar, where S is the start symbol.
- $S \rightarrow AC|CB$
 $A \rightarrow CA|a$
 $B \rightarrow AB|b$
 $C \rightarrow CB|b$
8. Construct a PDA for the given language $L = \{a^n b^n c^m | n, m \geq 1\}$ and show the moves made by the PDA for the following strings (a) $aabbbbcc$ (b) abc .
9. Construct a Turing machine for the language $L = \{a^n b^n c^n | n \geq 1\}$ and show the ID of the Turing machine of the string from abc to $a^2 b^2 c^2$.
10. Discuss any four types of TMs focusing on the transition function with a sample transition rule. Draw necessary diagrams at the required places.
11. Discuss
- Any two closure properties where recursive languages are closed
 - Any two closure properties where recursively enumerable languages are not closed.
12. Define Chomsky hierarchy of grammars with suitable examples and their corresponding machines.

$\Leftrightarrow \Leftrightarrow \Leftrightarrow H/E/TX \Leftrightarrow \Leftrightarrow \Leftrightarrow$

**VIT**

Vellore Institute of Technology

Final Assessment Test – April 2026

Course: BCSE304L - Theory of Computation

Class NBR(s): 3565 / 3588 / 3590

Slot: B1+TB1

Time: Three Hours

Max. Marks: 100

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COs	CO Statements
CO1	Compare and analyse different computational models.
CO2	Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO3	Identify limitations of some computational models and possible methods of proving them.
CO4	Represent the abstract concepts mathematically with notations.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions
(10 X 10 = 100 Marks)

1. Consider the languages $L_1 = \{a, ab, ba\}$ and $L_2 = \{b, aa\}$ defined over the alphabet $\Sigma = \{a, b\}$. Construct the languages obtained by performing the following operations: CO4 BL3
- $L_1 L_2$
 - $L_2 L_1$
 - L_1^3
 - $L_1 \cup L_2$
 - $L_1 \cap L_2$

Also, analyze the results obtained and determine which operation produces the maximum number of strings and which operation produces the minimum number of strings. Justify your answer based on the properties of language operations.

2. a) Consider the language L over the alphabet $\Sigma = \{0, 1\}$ such that, [5] CO2 BL3
 $L = \{w = t101 \mid t \in \Sigma^*\}$. That is, the language L consists of all binary strings (w) that end with 101. Construct the Deterministic Finite Automata (DFA) with transition table and identify the accepting state.

- b) Construct the Non-deterministic Finite Automata (NFA) over the [5]
 alphabet $\Sigma = \{a, b\}$ that accepts the language
 $L = \{w \mid w \text{ contains the substring "abba" or "bab"}\}$.

CO2 BL3

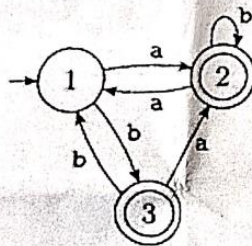
3. An NFA P is given by $(Q, \{0, 1\}, \delta, p_0, F)$ where $Q = \{p_0, p_1, p_2, p_3, p_4, p_5, p_6\}$,
 $F = \{p_4, p_6\}$ and the transition function δ is given below.

δ	0	1	ϵ
p_0	$\{p_0\}$	$\{p_0, p_1\}$	$\{p_2\}$
p_1	$\{p_3\}$	$\{p_2\}$	-
p_2	$\{p_4\}$	-	-
p_3	-	$\{p_5\}$	$\{p_4\}$
p_4	$\{p_4\}$	-	-
p_5	-	$\{p_6\}$	-
p_6	$\{p_3\}$	-	-

Find a DFA which is equivalent to the NFA P. Also say, whether the DFA is powerful than NFA. Justify your answer.

4. Given a finite automaton, evaluate its transition behaviour and systematically derive a regular expression that generates the same language. Discuss the steps involved and validate the correctness of the expression.

CO2 BL5



IS1 CO2 BL3

5. a) Convert the regular expression $(a + b)^*aba$ into Non-deterministic Finite Automata (NFA). [5] CO2 BL3

- b) Prove that the given language $L = \{a^{2n}b^na^n \mid n \geq 0\}$ is regular or not. [5]

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6. Given the grammar
 $S \rightarrow ABA \mid B \mid \epsilon$
 $A \rightarrow 00 \mid \epsilon$
 $B \rightarrow 11$

CO2 BL4

- a) Analyze the grammar and determine the nullable variables and unit productions. Explain how these productions influence the structure of the generated language.
 b) Transform the grammar into an equivalent grammar in Chomsky Normal Form (CNF) while preserving the language generated by the original grammar.
 c) Further convert the CNF grammar into Greibach Normal Form (GNF).

7. Construct a context-free grammar (CFG) for the language

CO3 BL6

$L = a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } j = k$. Construct the CFG for the language, explain how the grammar handles the two different conditions and provide the leftmost derivation and parse tree for a valid string.

8. Construct the pushdown automata (PDA) which accepts the language

CO3 BL6

$L = a^{2m}b^nc^na^{2m} \mid m, n \geq 0$. Give all the specifications of the PDA and the instantaneous description for any one of the strings accepted by the language.

- 9.a) Design of a Turing TM that accepts the language;

CO1 BL4

- i. $L = a^n b^{2n} \mid n \geq 1$ [5]

- ii. Analyse different types of Turing Machine (TM). [5]

- 9.a) Design of a Turing TM that accepts the language, CO1 BL4
- i. $L = a^n b^{2n} \mid n \geq 1$ [5]
- ii. Analyse different types of Turing Machine (TM). [5]

OR

- 9.b) Analyse the Turing Machine Halting Problem and discuss, with suitable scenarios, why it cannot be solved by any Turing Machine. CO1 BL4
- 10.a) Construct and explain the Chomsky Hierarchy by describing the different types of languages in order, their corresponding automata and the general form of their grammar rules. Analyse how the increase in computational power affects the types of languages that can be represented with suitable reasoning. CO1 BL4

OR

- 10.b) Critically analyze the Post Correspondence Problem and illustrate, with suitable examples, how it leads to undecidability in formal language theory. CO1 BL4

⇔⇔⇔ X/D/TZ ⇔⇔⇔